

Final Design Truss Lab Report

“Board of Trustees”

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Section: A2

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Department of Engineering
At
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Introduction

The goal of this project was to design a truss that can support at least two pounds of weight, given a specific set of restrictions. From the initial designs created, we narrowed our choice down to the one that supported a larger amount of weight and made improvements on it. The primary motivation for the final design was to improve on the load-to-cost ratio of our truss. If we were to develop a life-size truss modeled after the one developed in this project, we would want to make sure it has maximum efficiency given a lower budget. The initial design had a decent load-to-cost ratio, but when doing analysis on our work, we realized that some of the members and joints used did not really contribute to the overall load support of the design. Our final design supports a similar load, but costs less to make because it has less members and joints. In other words, it has a higher load-to-cost ratio. Another thing to note is that, since we prioritized increasing our load-to-cost ratio, we ended up disregarding the measured uncertainty. This allowed us to completely devote our attention to the max load; increasing that would increase the ratio that we wanted.

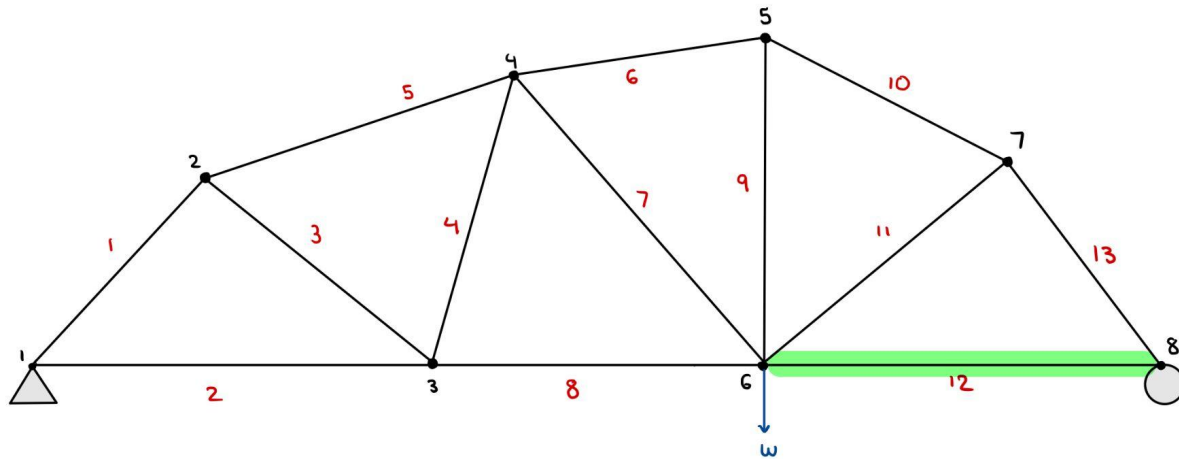
Procedure

The key change we made when creating the Final Design was removing all of the zero force members from the preliminary Design Report. This change was instrumental in reducing cost by limiting the number and total length of the members used in the truss design. Lowering cost, significantly increasing the max load to cost ratio. In other words, locating the zero force members and subsequently removing them, made our truss design far more cost efficient.

Analysis

As mentioned, the main change/difference between the Final Design and the Preliminary Analysis Report is the removal of the zero force members. Analysis from the preliminary report calculated the force experienced by each member of the truss, and thus identified the 2 zero force members. Since the goal of the Final Design was to maximize load to cost ratio, the zero force members were removed from the Truss in the Final Design. The removal of these zero force members leads to less material being used for the members which leads to less cost. Decreasing cost, increases the max load to cost ratio, satisfying our goal and motivation. From the code, we found that the uncertainty of the data was approximately ± 2.1 oz. However, we did not do anything to try and minimize this quantity because we wanted to focus on maximizing the maximum load of the truss.

Results



Note: The truss member highlighted in green (member 12) is predicted to buckle first

Member Number	Member Length (in)	Tension/Compression	Buckling Strength (oz)	Internal Force during max Load (oz)
1	8.1394	Compression	50.4	21.258 (C)
2	12	Tension	22.1	14.365 (T)
3	8.8459	Tension	42.2	10.782 (T)
4	9.1241	Compression	39.5	7.414 (C)
5	8.544	Compression	45.4	23.803 (C)
6	8.6313	Compression	44.5	33.341 (C)
7	12.3794	Tension	20.7	13.585 (T)
8	10	Tension	32.5	23.506 (T)
9	10.5	Tension	29.3	24.556 (T)
10	8.0623	Compression	51.4	37.816 (C)
11	9.5525	Tension	35.9	14.648 (T)
12	12	Tension	22.1 ±2.1 oz	22.100 (T)
13	8.2006	Compression	49.6	36.246 (C)

Summary:

- Max Load = 44.400 oz
- Theoretical max load/cost ratio = 0.2156 oz/\$
- Cost of truss: \$205.98

Discussion/Conclusion

Our team's primary goal was to design a truss that could potentially be used in real life. We wanted to simulate a truss that both supports a lot of weight and also can be produced for a reasonable cost. This is why our group decided to prioritize load-to-cost ratio. From the preliminary design, we saw that several members and joints were unnecessary; they did not contribute much to the overall strength of the truss. We particularly found that two of the members from the preliminary design were zero-force members. We kept them in the design at the time because we were well under our budget. However, for the final, we wanted to create a design that was as low-cost as possible. Therefore, to optimize the truss design, we removed the two zero-force members along with the joint that connected them together. Another thing to note is that, initially, we hoped to create a truss that was symmetrical so that it would support the same load at all parts and would also be appealing to the eye. However, in our final design, each member has a unique and odd length. This is because we realized that the criteria require us to support a load that is not at the center, but rather, on one side of the truss. Our design evolved over time to support more force on that side.

The most insightful part of this project was the MATLAB code used to simulate the truss. It makes sense to use a simulation as opposed to doing live testing because it saves time, materials, and money. That being said, one thing that would help improve the quality of the project is a little more support and access to materials during the actual building process. Since a lot of people were designing trusses at the same time, space and building materials, including tape, acrylic, and cutters were limited. Overall, though, the project was designed really well and helped us put our classroom knowledge to good use.

Appendix

Hartford Roof Collapse

Date: 26 April 2023

Start Time: 6:15 p.m.

Minutes of Meeting: 53 Minutes

Location: Warren Dormitory

Participants

- Anish Sinha: *Acting Chair*
- Arnav Mankad
- Jayden Raphino: *Recorder*

Agenda

1. Introduction of all team members
2. Discuss the Hartford Roof Collapse case
3. Pay attention to the use of computer programs in analysis and design
4. Determine an appropriate safety factor to be used in our own truss design
5. Report what maximum load we would claim for our design

Important Points

- **(Arnav Mankad):** The roof, being designed by Fraoli, Blum, and Yesselman, Engineers (F,B,&Y), consisted of a space frame composed of two layers of horizontal steel bars connected by diagonal bars. This design swayed away from the traditional space frame roof designing procedures in different ways. These include the use of a cross-shaped built up section for steel angles, which didn't provide great resistance to buckling, and the use of short posts on the nodes of the top layer in efforts to support the roofing panels instead of the top layer itself.
- **(Jayden Raphino):** To save money while designing, F,B,&Y used a state-of-the-art computer analysis to verify the safety and security of the building. However, despite said analysis predicted a downward deflection of 13 inches at the midpoint of the roof and an upward deflection of 6 inches at the corners, the actual deflection was measured to be twice as big as what was predicted. Despite this apparent discrepancy, F,B,&Y expressed no concern and proceeded construction.
- **(Anish Sinha):** As we know, the use of computer programs in analysis and design is common practice as it allows for more accurate and efficient calculations and simulations. However, as we saw throughout this report, computer analysis is only as good as what inputs you feed into the program. Also, variability in materials and

construction may also affect the program's accuracy, as well as unforeseen factors such as weather conditions.

- **(Arnav and Anish):** An appropriate safety factor for a truss design in a case where human life is at risk should be determined based on the specific application and use of whatever you're building, in this case a truss. So, a safety factor of at least 1.5 is commonly used when accounting for uncertainties in the design process. But a higher safety factor should be good in cases where human life is at risk.
- **(Jayden Raphino):** In order to determine the maximum load that can be claimed for a truss design, the design must consider its intended use and the maximum loads it may be subjected to. For example, a truss designed for a pedestrian bridge would have different load requirements than an overpass. The design must also consider the variability in materials/construction, as well as factors such as weather conditions, seismic activity, and other environmental factors.

Conclusion: The unfortunate incident of the Hartford Roof Collapse serves as a reminder of the importance of Fundamental Canon No.1 from the Code of Ethics of Engineers, National Society of Professional Engineers, 2018, which states that "*Engineers shall hold paramount the safety, health, and welfare of the public.*" This statement is critically important for engineers to prioritize safety on their designs and to constantly reassess and improve their designs based on new information and developments, otherwise drastic consequences would be present as a result of not adhering to it.